

FBPIC LWFA Baseline: Diagnostic Analysis

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FBPIC Baseline Simulation and Initial Analysis

Purpose

This is a baseline run before the main MPhys project. The aim is to get a working FBPIC and openPMD workflow, with detailed analysis and clear, concise figures produced. This should demonstrate the key features of laser wakefield acceleration, such as a low-density electron depletion zone with a high-density sheath. It's important to note that this is only a preliminary learning and validation run, not an optimisation project.

Computing Environment

Computing Element	Information
System	Windows laptop
Linux environment	Ubuntu through WSL
Editor	Visual Studio Code connected to Ubuntu
Python environment	Conda environment: fbpic
Simulation code	FBPIC
Diagnostic reader	openPMD-viewer
Hardware used for this run	Intel Core Ultra 5 115U CPU; integrated Intel graphics not used for CUDA acco

Baseline Simulation

Quantity	Baseline value
Simulation script	simulation/lwfa_baseline.py
Total iterations	1,800
Diagnostic interval	50 iterations
Saved outputs	36 snapshots, iterations 0-1750
Final analysed iteration	1750
Final analysed time	291.9 fs
Reconstructed E_z grid	100 × 800 cells

Exact Data

Parameter	Symbol	Value (exactly as written in script)
Plasma electron density	n_e	4.e18*1.e6
Normalised laser amplitude	a_0	4.
Laser waist	w_0	5.e-6
Pulse duration	τ	16.e-15
Laser wavelength	Inherent quantity of GaussianLaser	0.8e-6

Parameter	Symbol	Value (exactly as written in script)
Longitudinal resolution	N_z	800
Radial resolution	N_r	50
Azimuthal modes	N_m	2
Macroparticle sampling	(p_{nz}, p_{nr}, p_{nt})	(2, 2, 4)
Density-ramp length	ramp_length	40.e-6

Equations and theory

For an electron plasma density n_e , the plasma angular frequency is:

$$\omega_p = \sqrt{\frac{n_e e^2}{m_e \epsilon_0}}.$$

The corresponding plasma wavelength is:

$$\lambda_p = \frac{2\pi c}{\omega_p}.$$

The cold plasma wave-breaking field scale is:

$$E_0 = \frac{m_e c \omega_p}{e}.$$

These are directly calculated using the density from the analysis code rather than copying the reference values. I calculated these directly as an additional sanity check.

For electron plasma density $n_e = 4.0 \times 10^{24} \text{ m}^{-3}$: |Quantity|Calculated value| |:-:| |Plasma angular frequency, ω_p | $1.1283 \times 10^{14} \text{ rad s}^{-1}$ | |Plasma wavelength, λ_p | $16.6947 \text{ }\mu\text{m}$ | |Cold wave-breaking field scale, E_0 | $192.318 \text{ GV m}^{-1}$ |

E_0 is a reference scale, not a strict upper limit on the field, as in this simulation it is expected there will be strongly nonlinear wakes.

Diagnostic 1: two-dimensional longitudinal field

To verify that a clear longitudinal wake had formed, a standard two-dimensional E_z plot was mapped. This additionally showed where the strongest field regions were, guiding future diagnostics.

The longitudinal electric field was loaded with `OpenPMDTimeSeries("diags/hdf5")`, which reconstructed the $\theta = 0$ plane and converted the field from V m^{-1} to GV m^{-1} before plotting $E_z(r, z)$.

At iteration 1,750, the full reconstructed field had positive and negative extremes of:

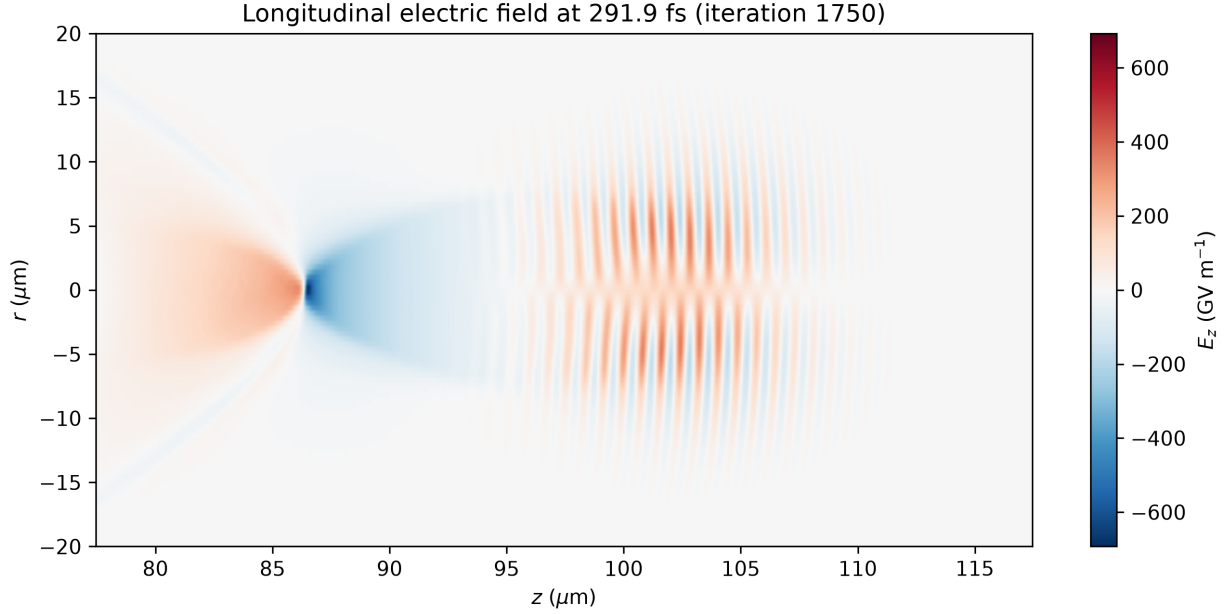
$$E_{z,\min} \approx -693 \text{ GV m}^{-1}, \quad E_{z,\max} \approx +356 \text{ GV m}^{-1}.$$

The main wake can be clearly seen near the propagation axis. There is also a shorter-scale anti-symmetric structure farther off-axis, which is expected to cancel during on-axis analysis. Subsequently, these oscillations were not treated as a direct measurement of the plasma wavelength.

As for electrons in the field, $F_z = -eE_z$, a negative E_z corresponds to a force in the positive- z direction. This assumes positive- z is the forward propagation direction.

Associated files:

- [analysis/plot_ez.py](#)
- [figures/ez_final_full_scale.png](#)



Diagnostic 2: on-axis longitudinal field

The field was then reduced to an on-axis lineout in order to compare extrema, their positions and the longitudinal wake shape quantitatively.

The reconstructed slice contained the two cells immediately adjacent to $r = 0$, with their average used to estimate the on-axis field:

$$E_{z,\text{axis}}(z) = \frac{1}{2}[E_z(r_-, z) + E_z(r_+, z)].$$

The resulting lineout gave:

Measurement	Result
Maximum on-axis field	+323.86 GV m ⁻¹
Position of maximum	86.03 μm
Minimum on-axis field	-679.76 GV m ⁻¹
Position of minimum	86.53 μm

This showed the minimum on-axis longitudinal field to be:

$$E_{z,\text{min}} \approx -680 \text{ GV m}^{-1}.$$

This is significant when compared with the cold wave-breaking field scale, $E_0 \approx 192.3 \text{ GV m}^{-1}$, as the local minimum on-axis longitudinal field is much larger than this value.

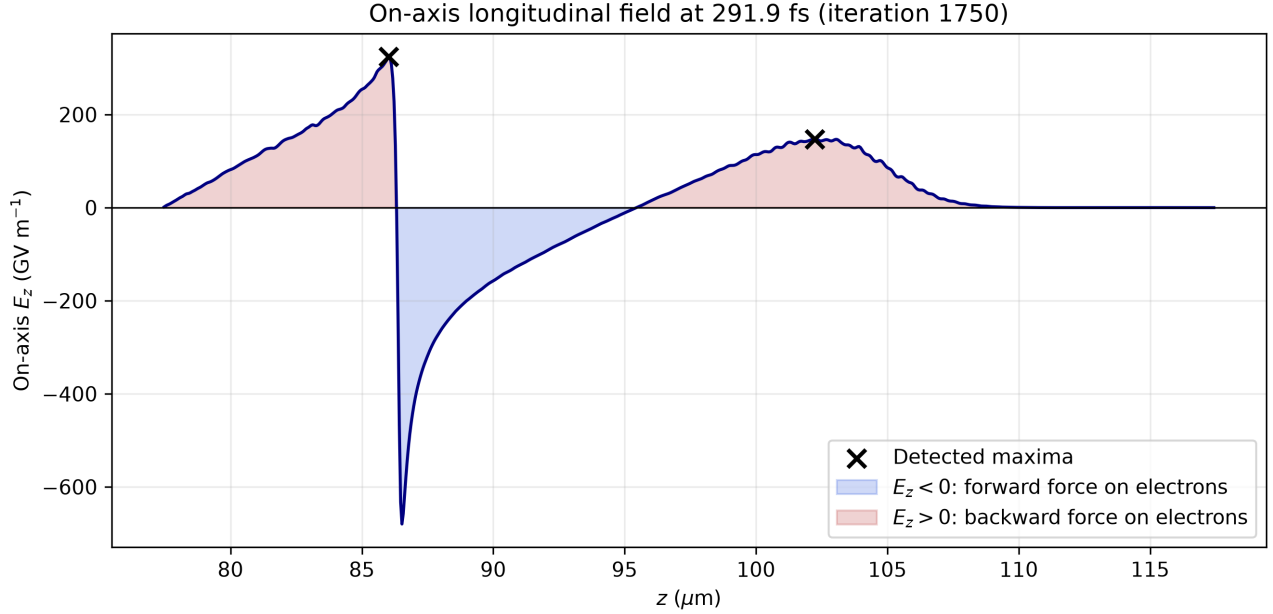
Therefore the local minimum is approximately:

$$|E_{z,\text{min}}| \approx 3.5 E_0.$$

Importantly, the fact that the minimum on-axis field is about $3.5E_0$ does not automatically mean something is wrong, because E_0 is the characteristic cold-plasma scale, and this simulation deals with strongly non-linear wakes. However, this value alone doesn't prove that a strong non-linear wake was simulated. The feature will be investigated in further diagnostics, before taking -680 GV m^{-1} as a representative wake amplitude.

Associated files:

- [analysis/plot_ez_lineout.py](#)
- [figures/ez_final_on_axis.png](#)



Diagnostic 3: preliminary wake-scale measurement

This diagnostic aimed to measure the separation between strong positive E_z peaks as a simple sanity check against the theoretical plasma wavelength.

Positive local maxima were detected using `scipy.signal.find_peaks`. A minimum prominence of 50 GV m^{-1} was used to reject small ripples, while the minimum allowed peak separation was decided using the expected plasma wavelength and longitudinal grid spacing.

Detected positive maxima:

Peak	Position, z	Field, E_z
1	86.025 μm	323.86 GV m^{-1}
2	102.225 μm	147.40 GV m^{-1}

The measured separation was

$$\Delta z = 102.225 \mu\text{m} - 86.025 \mu\text{m} = 16.200 \mu\text{m},$$

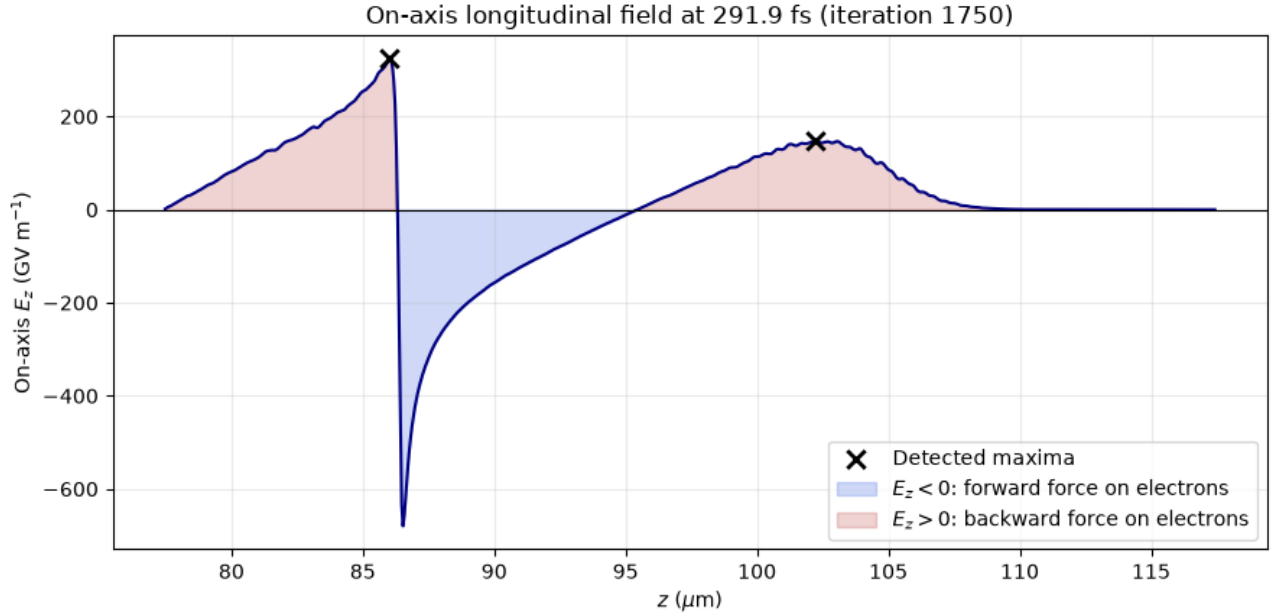
which is approximately 3.0% shorter than

$$\lambda_p = 16.6947 \mu\text{m}.$$

However, despite this result looking encouraging, the later E_x and laser-envelope plots placed the second maximum ($z = 102.225 \mu\text{m}$) inside the main laser region, strongly suggesting that these peaks were not a clean measurement of successive wake periods. This agreement is still a useful scale check, but to perform a proper wavelength measurement, the wake would need to be isolated behind the driver, and utilise more than one interval.

Associated files:

- [analysis/measure_wake_wavelength.py](#)
- [figures/measure_wake_wavelength.png](#)



Diagnostic 4.1: two-dimensional electron density

To further verify whether a wake was produced, the reconstructed electron density was next analysed to see what plasma structure sits around the large E_z feature. The charge density was loaded at $\theta = 0$ and converted to an electron-density estimate using:

$$n_e = -\frac{\rho}{e}.$$

This was then normalised to the baseline density

$$n_0 = 4.0 \times 10^{24} \text{ m}^{-3}.$$

At iteration 1,750, the reconstructed plane gave:

Measurement	Result
Minimum n_e/n_0	-0.111
Maximum n_e/n_0	127.558
Median n_e/n_0	0.999
99th percentile n_e/n_0	2.380

On this full scale plot, almost all of the colour range is taken up by a very narrow on-axis density compression close to

$$z \approx 86.5 \mu\text{m}.$$

This is the same position as the large negative E_z spike, which is consistent with a wake, however, not conclusive. The median remains almost exactly at the background density, so the $127.6n_0$ value is clearly a very local feature rather than something representative of the whole plane.

Diagnostic 4.2: scaled two-dimensional electron density

To show the ordinary wake structure more clearly, a second density plot was produced with the colour scale clipped to

$$0 \leq \frac{n_e}{n_0} \leq 5.$$

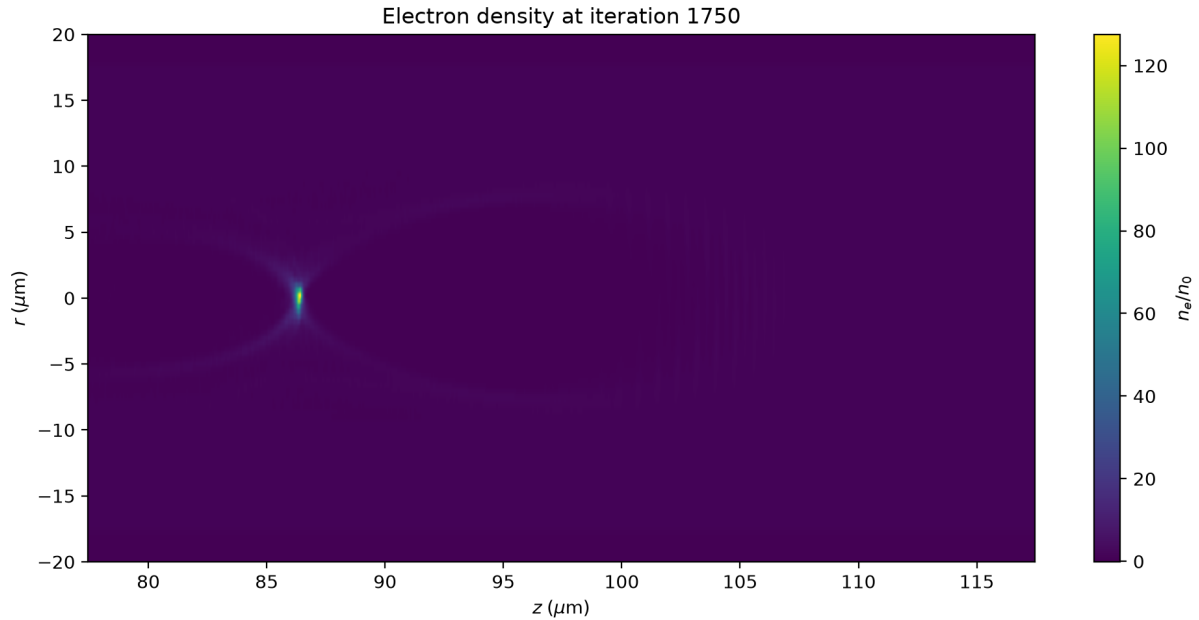
The data was left completely unchanged; only the displayed colour range is different. This makes the depleted on-axis cavity and the enhanced-density sheath much easier to see. The sheath converges near $z \approx 86.5 \mu\text{m}$, giving a structure that looks consistent with a strongly nonlinear, blowout-like wake behind the laser.

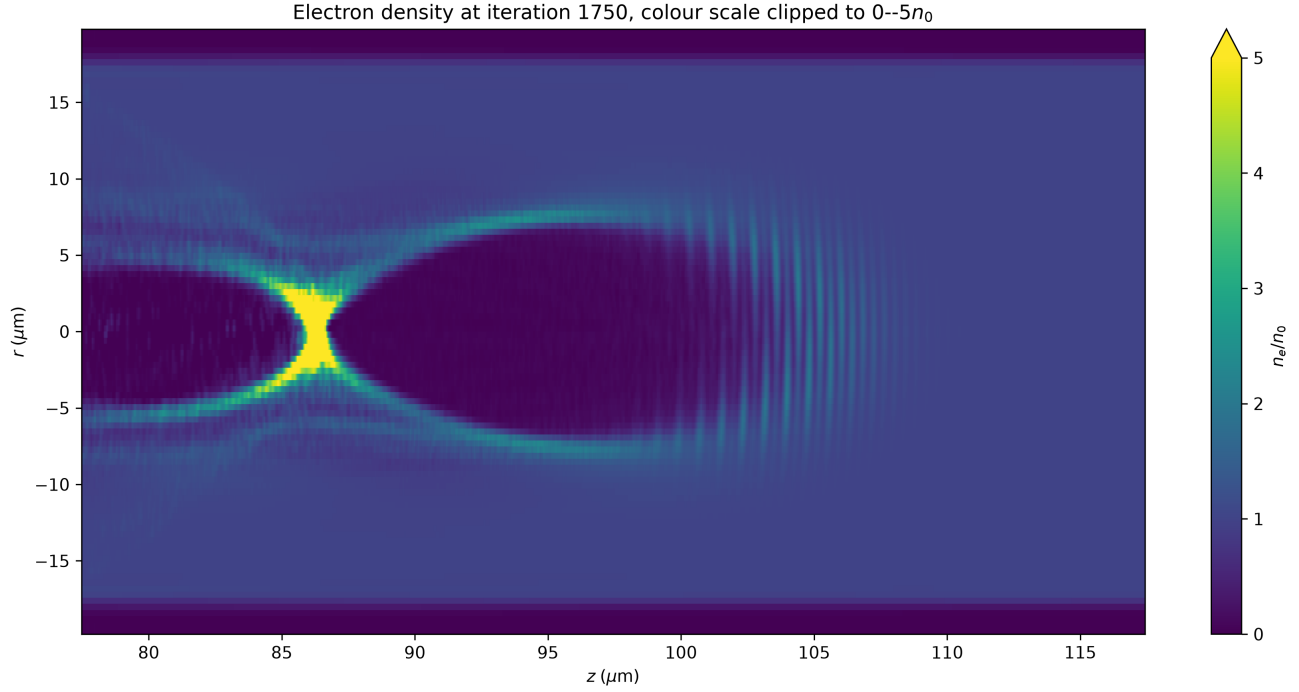
Important diagnostic note

The small negative minimum recorded is not physical. It is treated as a numerical reconstruction effect around a very sharp feature, with finite resolution and the limited azimuthal-mode representation both possible contributors. For the same reason, I would not treat the exact $127.6n_0$ maximum as truly converged without further resolution and mode checks.

Associated files:

- [analysis/plot_ne.py](#)
- [analysis/plot_density_clipped.py](#)
- [figures/electron_density_final_full_scale.png](#)
- [figures/electron_density_final_clipped_0_5n0.png](#)





Diagnostic 5: two-dimensional transverse laser field

The electric field in x -direction, E_x , was plotted to locate the laser relative to the wake and to check that the transverse pulse and its rapid carrier oscillations were still clearly resolved at the iteration when the final snapshot was taken.

The x -component of the electric field was reconstructed at $\theta = 0$, converted from V m^{-1} to TV m^{-1} and plotted as $E_x(r, z)$. At iteration 1,750, this produced the field extrema below:

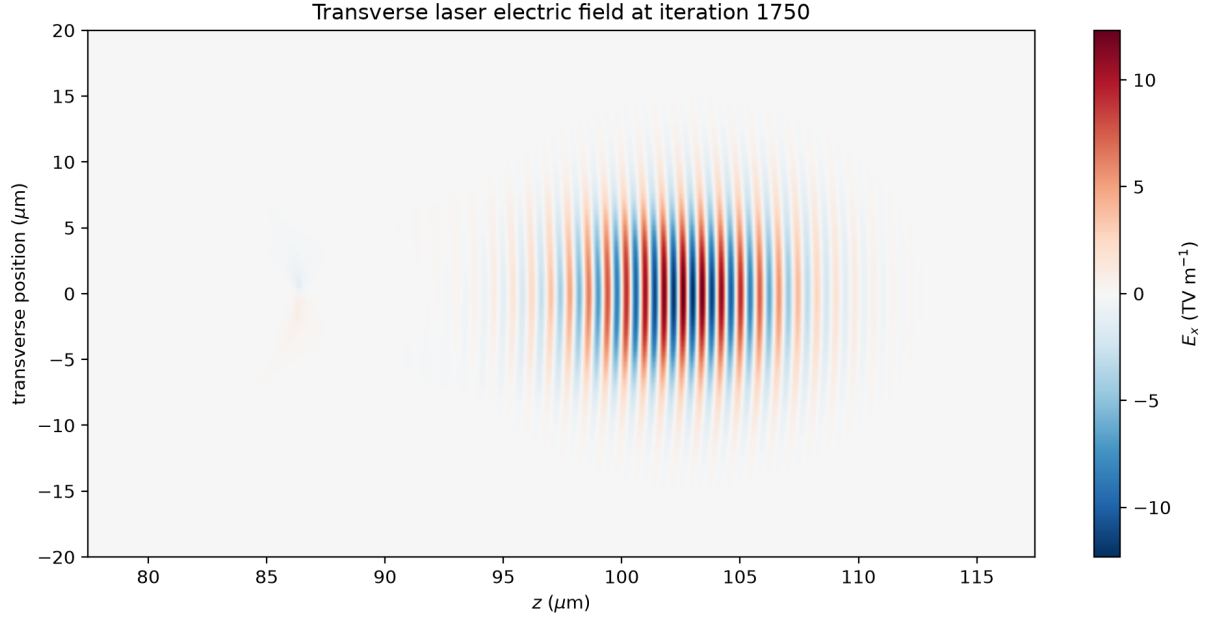
$$E_{x,\min} \approx -12.290 \text{ TV m}^{-1}, \quad E_{x,\max} \approx +12.320 \text{ TV m}^{-1}.$$

The plot showed the main laser pulse to be centred about $z \approx 103 \mu\text{m}$. This is well ahead of the strong density and E_z feature near $z \approx 86.5 \mu\text{m}$. This is the expected ordering if the $z \approx 86.5 \mu\text{m}$ feature belongs to a wake behind the driving laser.

The raw E_x plot still contains the fast optical carrier, so I did not use a single field maximum as the laser position. I extracted the envelope later and used that to track the pulse more consistently.

Associated files:

- [analysis/plot_ex.py](#)
- [figures/ex_final_full_scale.png](#)



Diagnostic 6: longitudinal electron phase space

To further verify that a useful wake had been created, the next diagnostic would test whether any distinct high-momentum electron population actually developed. This could be shown by plotting longitudinal position against u_z .

The saved electron positions and normalised longitudinal momenta were loaded using `var_list=["z", "uz"]`, where

$$u_z = \frac{p_z}{m_e c}.$$

At iteration 1,750, the saved-particle distribution gave:

Measurement	Result
Saved macroparticles	50,357
Longitudinal range	72.650-106.081 μm
Minimum u_z	1.000
Maximum u_z	11.905
Median u_z	1.586
95th percentile u_z	4.066
99th percentile u_z	8.189

This produced two visibly different structures: a high-momentum branch sitting around $z \approx 84$ - $87 \mu\text{m}$ and reaching $u_z \approx 11.9$, and a more regular lower-momentum oscillatory pattern which appeared around $z \approx 101$ - $106 \mu\text{m}$. This further supports previous conclusions, with the E_x plot placing the laser in the second region, suggesting the oscillatory population is laser-associated, whereas the high-momentum branch is consistent with a wake.

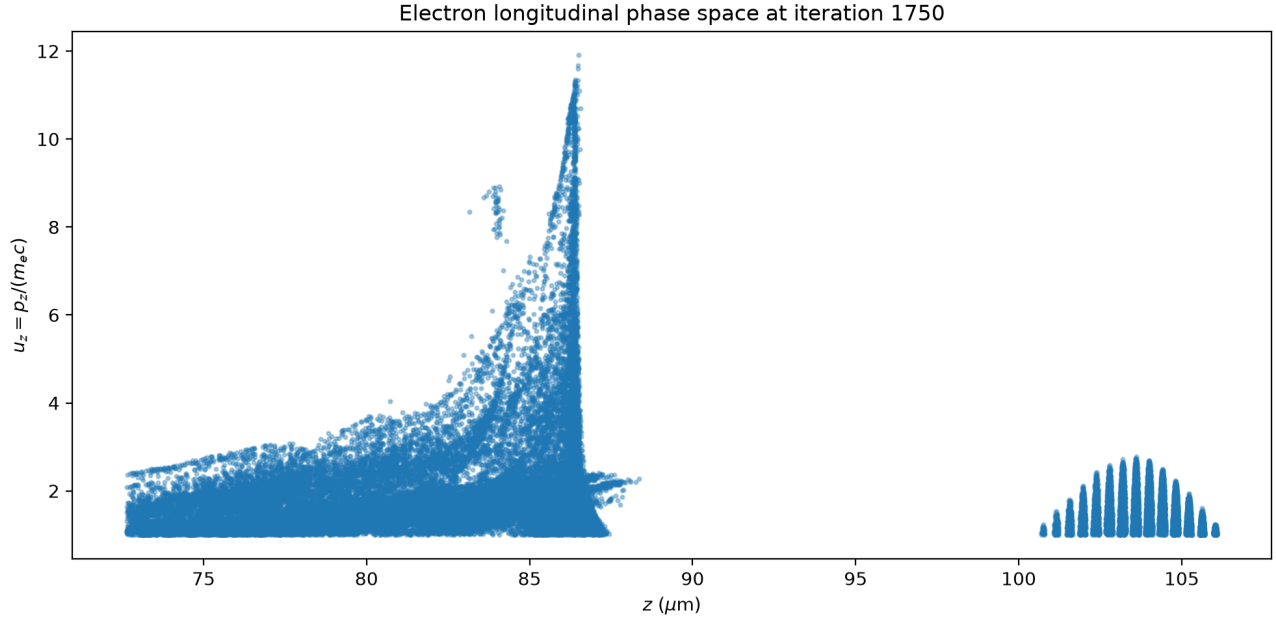
Diagnostic note

One limitation created by the saved particle data was the diagnostic only keeping electrons with `select={"uz": [1., None]}`. Therefore, anything below $u_z = 1$ is missing from the plot, so the

plot is not fully representative of the complete plasma distribution. The scatter density is also only macroparticle density, with the points not weighted by the physical number of electrons they represent.

Associated files:

- [analysis/plot_z_uz.py](#)
- [figures/z_uz_phase_space_final.png](#)



Diagnostic 7: weighted electron energy spectrum of the saved population

Logically, the next step was to convert the saved particle momenta into kinetic energy and include the macroparticle weights so that the spectrum represented physical electron charge rather than just computational particle counts.

The Lorentz factor and kinetic energy were calculated from the three normalised momentum components:

$$\gamma = \sqrt{1 + u_x^2 + u_y^2 + u_z^2}, \quad E_k = (\gamma - 1)m_e c^2.$$

At iteration 1,750, the weighted distribution gave:

Measurement	Result
Saved macroparticles	50,357
Minimum kinetic energy	0.214 MeV
Maximum kinetic energy	5.611 MeV
Weighted mean energy	1.122 MeV
Weighted RMS spread	0.965 MeV
Charge of saved $u_z \geq 1$ population	346.916 pC

The spectrum produced was broad and mostly weighted towards lower energies, with a tail reaching to about 5.6 MeV. Importantly, this cannot be called a true "beam spectrum", as the $u_z \geq 1$ diagnostic mixes several different forward-moving populations together.

Macroparticle weights were included through

$$Q = e \sum_i w_i,$$

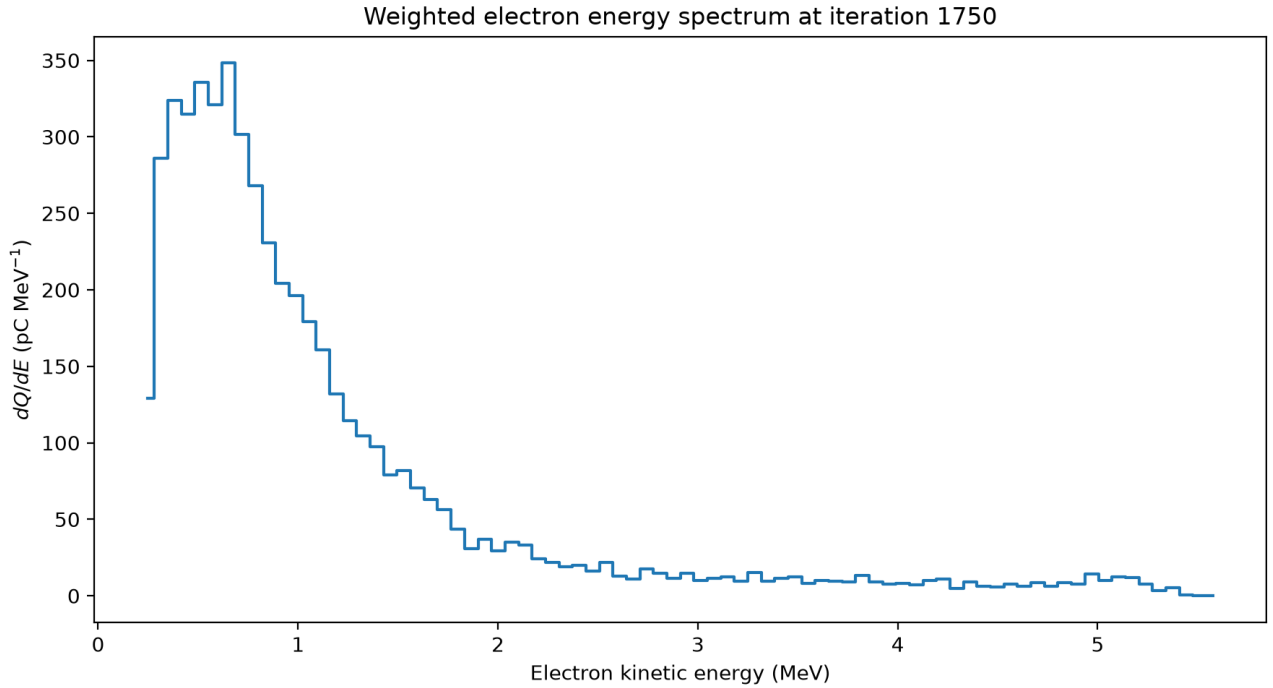
which gave

$$Q_{\text{saved}} = 346.916 \text{ pC}.$$

The charge Q_{saved} represents the charge of every saved electron above the diagnostic cutoff, whether or not that electron belongs to the late wake-associated population.

Associated files:

- [analysis/plot_energy_spectrum.py](#)
- [figures/electron_energy_spectrum_final.png](#)



Diagnostic 8: maximum saved-electron energy versus time

The next key diagnostic was to calculate the maximum energy over all 36 saved snapshots to see both when high-energy particles first appeared, and whether the final 5.611 MeV value developed smoothly.

The methodology was to repeat the kinetic energy calculation from Diagnostic 7 for all 36 saved iterations. This didn't produce any particles satisfying the $u_z \geq 1$ diagnostic selection for the first five saved outputs, with the first saved $u_z \geq 1$ particles appearing at approximately 41.7 fs.

When kinetic energy was calculated for each iteration, the maximum saved-electron energy then showed a strongly non-monotonic evolution:

Time	Maximum kinetic energy
41.70 fs	1.139 MeV
83.39 fs	4.671 MeV
150.10 fs	2.380 MeV

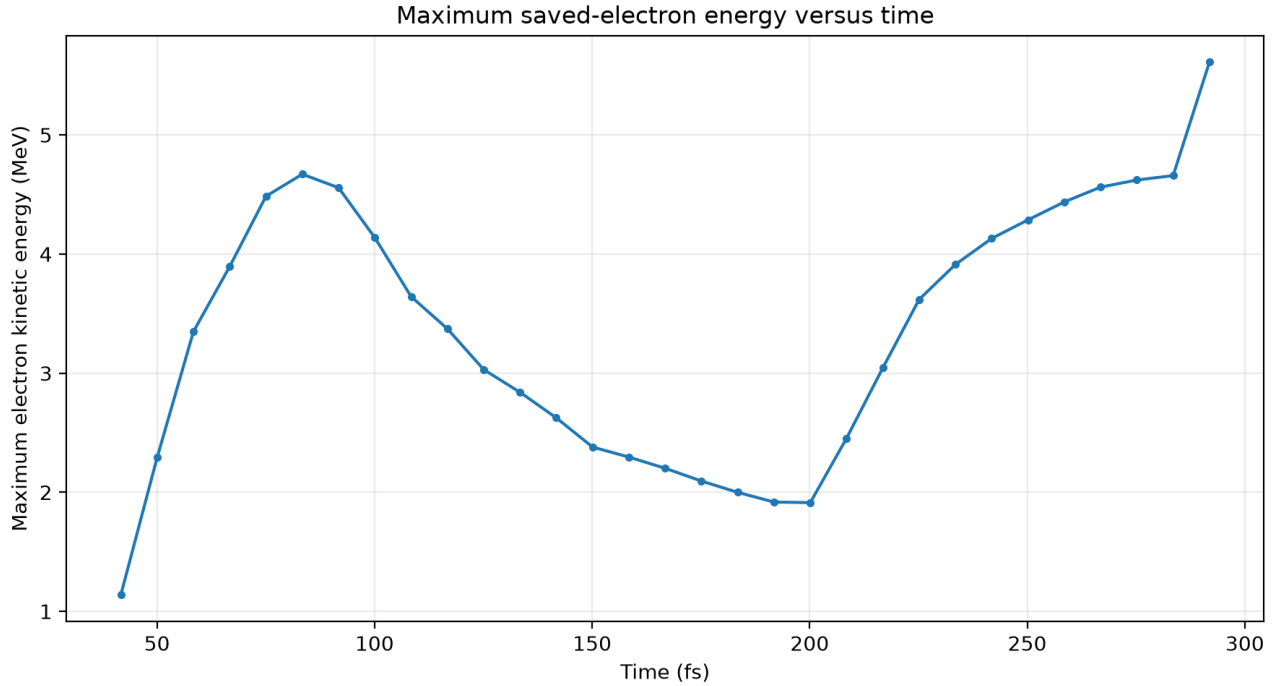
Time	Maximum kinetic energy
200.14 fs	1.913 MeV
250.17 fs	4.287 MeV
291.87 fs	5.611 MeV

The above graph demonstrates the maximum rising quickly, then falling throughout the middle of the run and then rising again after about 200 fs. This does not mean one electron accelerated, lost energy and then accelerated again. Any macroparticle can be the maximum at each snapshot point.

The later phase-space and laser-envelope plots make this much clearer, with the early high-energy feature overlapping with the laser region, while the final high-momentum population is clearly behind the driving laser. This is a good example of why $E_{\max}$ on its own is a poor optimisation metric.

Associated files:

- [analysis/plot_max_energy_vs_time.py](#)
- [figures/max_electron_energy_vs_time.png](#)



Diagnostic 9: longitudinal-field extrema versus time

Following the unexpected maximum energy result, $E_{\max}$ was subsequently compared with the on-axis E_z extrema over the same saved iterations. This would investigate if a link existed between the strongly non-monotonic evolution of $E_{\max}$, and the evolution of the on-axis field E_z .

The on-axis E_z lineout was reconstructed at every saved iteration with the positive and negative extrema recorded. The longitudinal wake grew strongly through the simulation, reaching the final values found in the earlier diagnostic which only examined iteration 1750 of:

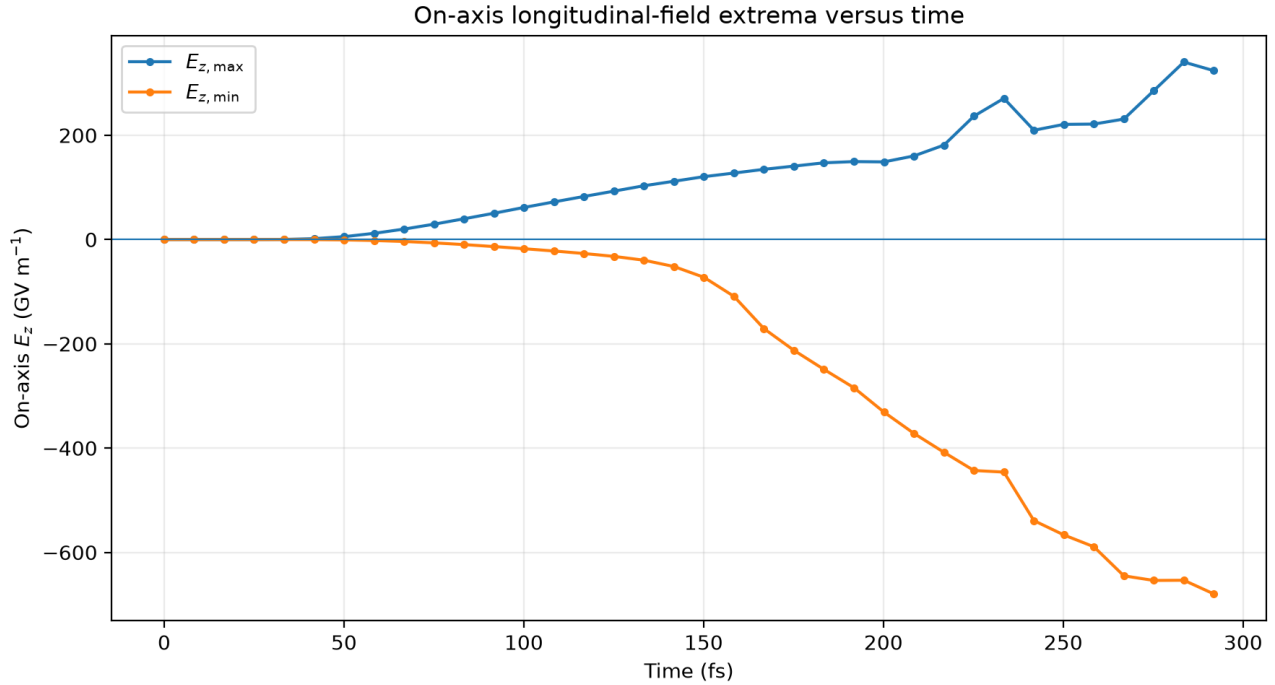
$$E_{z,\min} \approx -680 \text{ GV m}^{-1}, \quad E_{z,\max} \approx +324 \text{ GV m}^{-1}.$$

Interestingly, between 80 and 200 fs, the negative E_z extremum becomes much stronger while the maximum saved-electron energy falls. If maximum electron energy were mainly controlled by the peak wakefield, I would expect the two to increase together. Instead, this is not seen here. This supports the idea that the early high $E_{\max}$ values did not come from the wake, but instead from the laser.

After about 200 fs, E_z keeps strengthening and a separate high-momentum population develops behind the laser. The maximum energy rises again at the same time, which fits the late wake-associated interpretation much better than the early energy peak.

Associated files:

- [analysis/plot_ez_extrema_vs_time.py](#)
- [figures/ez_extrema_vs_time.png](#)



Diagnostic 10: longitudinal phase-space evolution

As a further test, the phase space was plotted at five times to see whether the rise-fall-rise in $E_{\max}$ came from one evolving structure or from different populations appearing during the run.

The z - u_z distribution was compared at five representative saved iterations:

Iteration	Time	Saved macroparticles	Maximum u_z
500	83.39 fs	28,172	9.053
900	150.10 fs	35,352	4.032
1200	200.14 fs	38,775	4.272
1500	250.17 fs	46,384	8.506
1750	291.87 fs	50,357	11.905

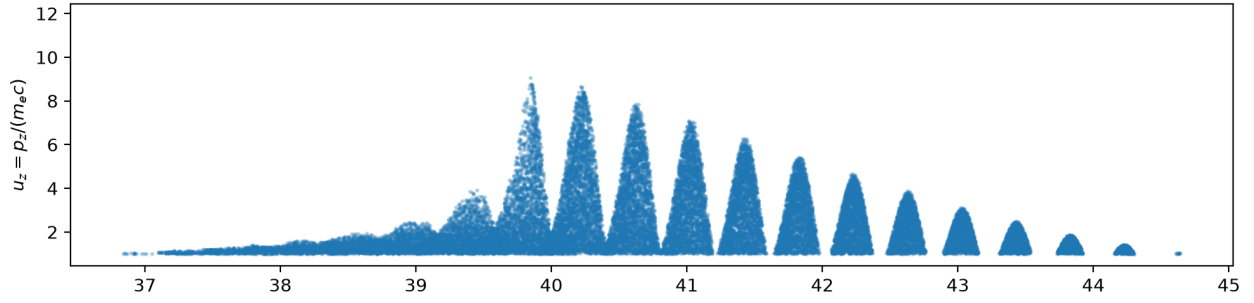
At 83.4 fs the largest u_z value is part of a regular sequence of peaks near the front of the saved distribution. The high-momentum feature mostly disappears by 150-200 fs, but by 250 fs, a different branch has appeared well behind the regular forward oscillations, and this later branch becomes much stronger by the final snapshot.

This is key: the plots do not look like one bunch losing and regaining energy; instead, the phase-space structure itself changes. To reasonably verify the theory that the irregular $E_{\max}$ values were caused by the laser, one last diagnostic is required to test whether the regular forward pattern was moving with the laser.

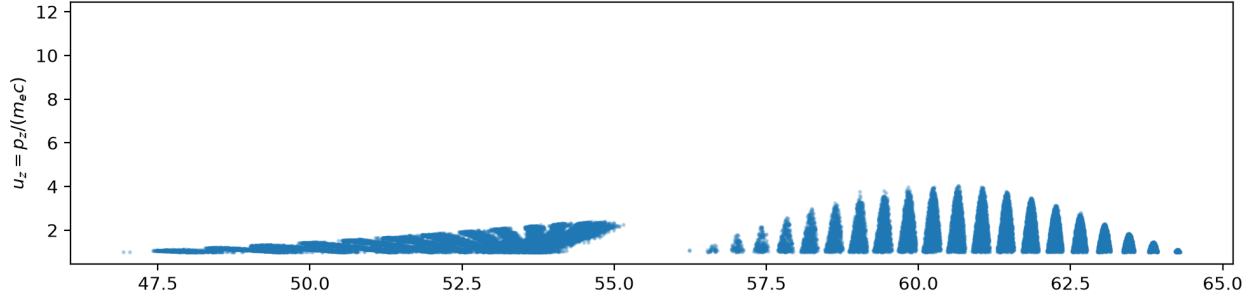
Associated files:

- [analysis/plot_z_uz_evolution.py](#)
- [figures/z_uz_phase_space_evolution.png](#)

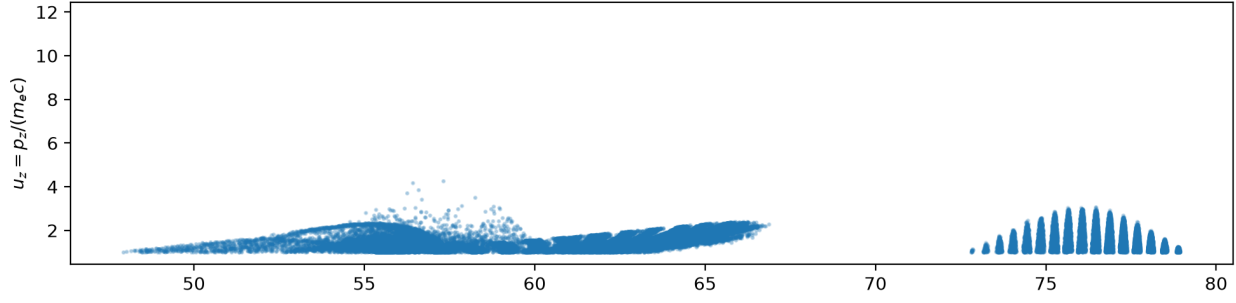
Evolution of electron longitudinal phase space ($u_z \geq 1$ saved population)
Iteration 500 — 83.4 fs



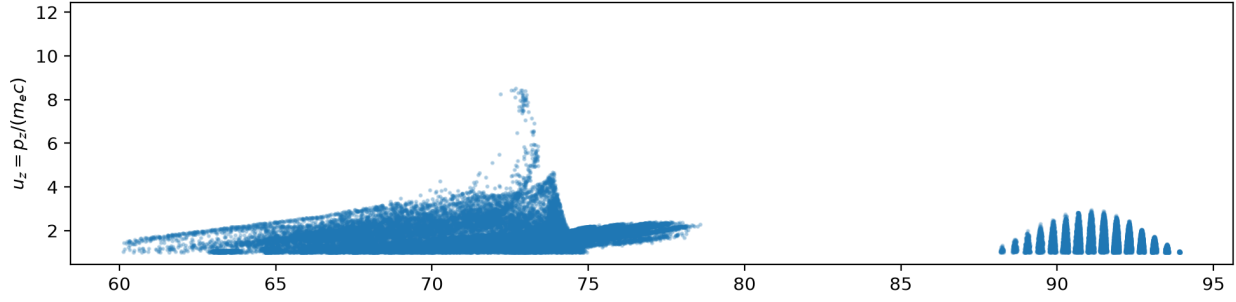
Iteration 900 — 150.1 fs



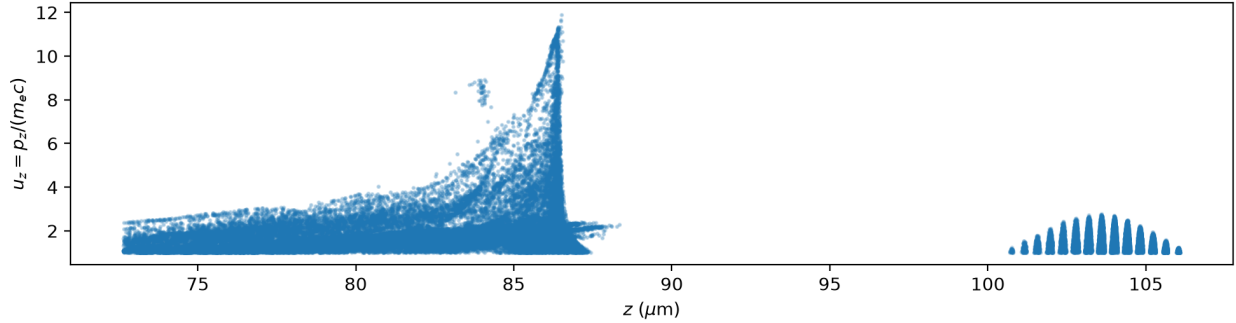
Iteration 1200 — 200.1 fs



Iteration 1500 — 250.2 fs



Iteration 1750 — 291.9 fs



Diagnostic 11: laser position and envelope evolution

To test if the regular forward phase-space pattern is tied to the laser rather than the trailing wake, the laser envelope can be tracked at the same times the phase-space pattern is seen.

For each selected iteration, the two cells nearest the axis were averaged to obtain an on-axis $E_x(z)$ lineout. The rapidly oscillating carrier was converted to an envelope estimate using the magnitude of the analytic signal from a Hilbert transform. The position of the largest envelope value was used as the laser-envelope peak position.

Iteration	Time	Laser-envelope peak z	Peak envelope
500	83.39 fs	39.931 μm	15.509 TV m^{-1}
900	150.10 fs	59.831 μm	14.559 TV m^{-1}
1200	200.14 fs	75.081 μm	13.490 TV m^{-1}
1500	250.17 fs	90.332 μm	12.850 TV m^{-1}
1750	291.87 fs	102.832 μm	12.512 TV m^{-1}

This shows a clear result: the laser-envelope peak moves with the regular peak-dip-peak pattern in the z - u_z plots. This is strong evidence to treat that forward oscillatory structure as laser-associated rather than part of the late wake bunch.

The peak envelope also falls from about 15.5 to 12.5 TV m^{-1} , around a 19% reduction. That shows the pulse is changing, however, this is not necessarily a 19% laser-energy loss. This is because a change in pulse width, focusing or shape can also change the peak field, so the evidence is insufficient.

At the final snapshot the laser-envelope maximum is at

$$z_{\text{laser}} = 102.832 \mu\text{m},$$

whereas the strong high-momentum trailing branch is around

$$z \approx 86.5 \mu\text{m}.$$

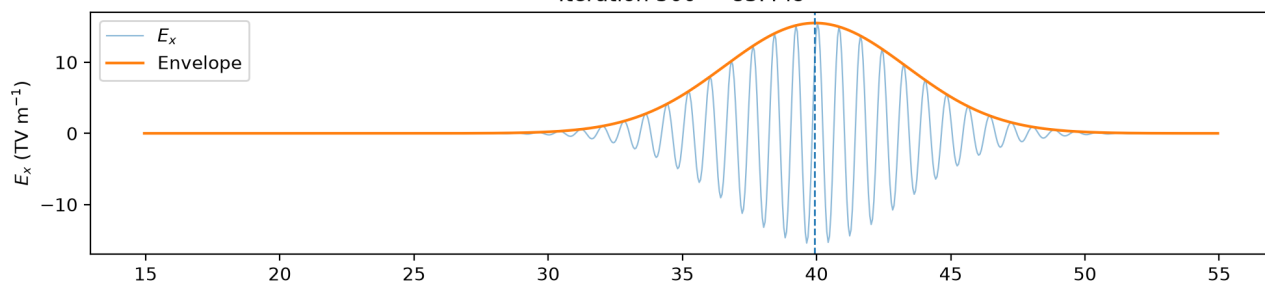
This is a separation of approximately 16.3 μm , close to the reference $\lambda_p = 16.6947 \mu\text{m}$. This puts the late high-momentum population at roughly the expected wake scale behind the driving laser. This is a geometrical consistency check, not a precision measurement of the nonlinear wake period.

Associated files:

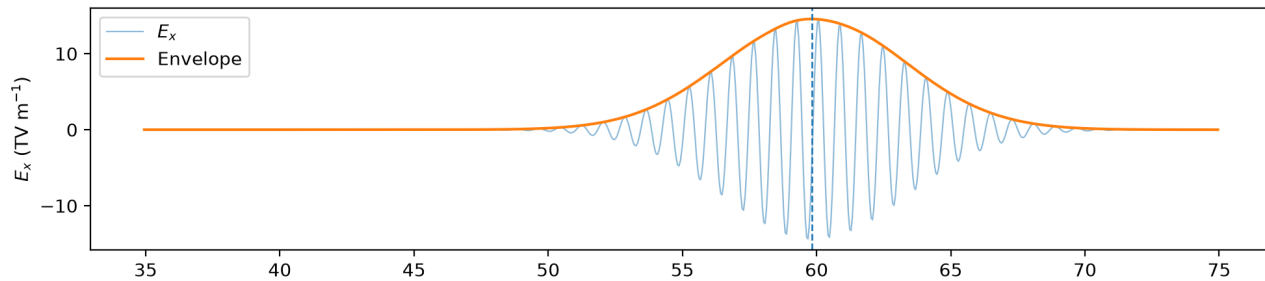
- [analysis/plot_laser_position_evolution.py](#)
- [figures/laser_position_evolution.png](#)

Evolution of the on-axis laser field and envelope

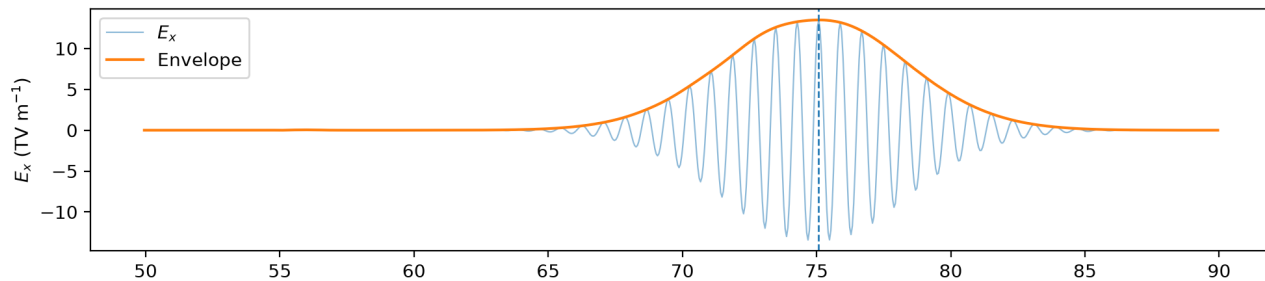
Iteration 500 — 83.4 fs



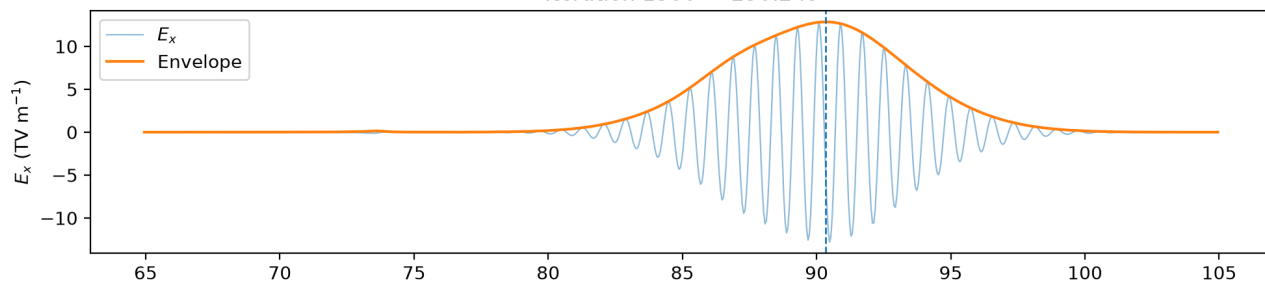
Iteration 900 — 150.1 fs



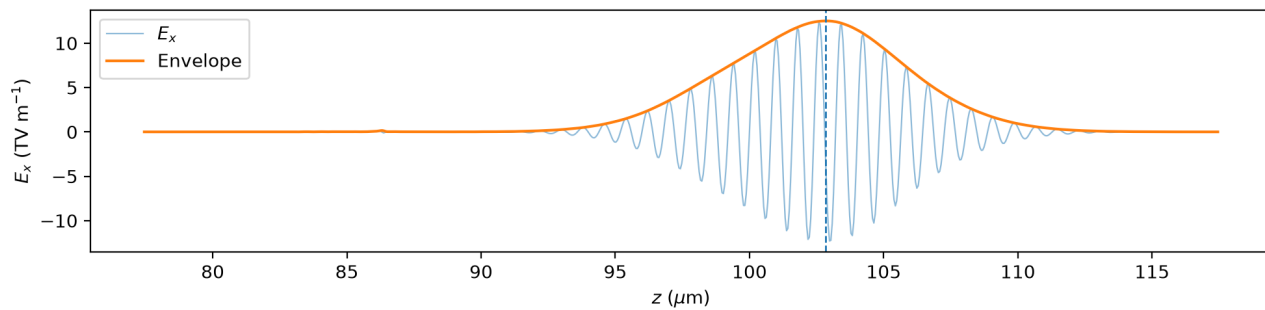
Iteration 1200 — 200.1 fs



Iteration 1500 — 250.2 fs



Iteration 1750 — 291.9 fs



Diagnostic 12: robustness of the large E_z spike

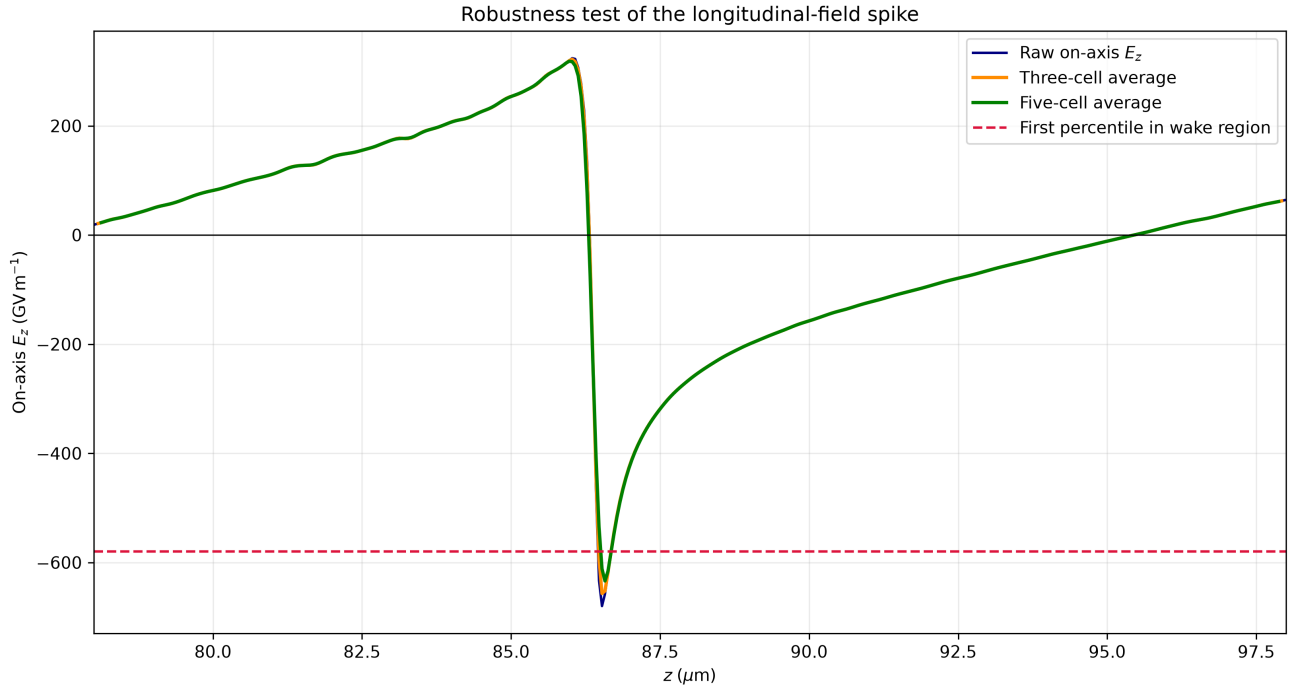
To test the validity of the large and very sharp -680 GV m^{-1} minimum, the raw on-axis lineout was compared with three-cell and five-cell moving averages around the same sharp feature. A first-percentile value was also calculated within the selected wake region to provide a measure that is not controlled by the single most extreme cell. The reliability of this measurement is strengthened if it can survive modest spatial averaging.

While the exact minimum becomes less negative after spatial averaging, the feature remains of order several hundred GV m^{-1} and the three- and five-cell averaged curves continue to reach below approximately -600 GV m^{-1} . The first percentile of the wake-region field is also strongly negative, at roughly -580 GV m^{-1} .

The feature survives both averages, so the strong negative field is not just one bad cell. The raw -680 GV m^{-1} value is still a very local extreme, though, and should be treated as a peak rather than as the typical wake amplitude. The fact that the feature survives modest averaging strengthens the case that it is not simply a single-cell numerical artefact.

Associated files:

- [analysis/test_ez_spike_robustness.py](#)
- [figures/ez_spike_robustness.png](#)



Diagnostic 13.1: candidate wake-bunch selection and beam metrics

Finally, the late high-momentum branch was separated from the full saved $u_z \geq 1$ population so that some quantities for a physically motivated candidate bunch could be calculated, rather than for every forward-moving electron.

A laser-relative coordinate was defined as

$$\xi = z - z_{\text{laser}},$$

where $\xi = 0$ corresponds to the final laser-envelope peak at $z_{\text{laser}} = 102.832 \mu\text{m}$. Negative ξ is therefore behind the laser.

Electron-bunch charge is represented as a positive charge magnitude and the candidate electron bunch was selected using

$$-20 \leq \xi \leq -14 \mu\text{m},$$

and

$$u_z \geq 4.$$

The spatial window was chosen to cover the late branch behind the laser, while $u_z \geq 4$ cuts away most of the large low-momentum population underneath it. As these are analysis choices and not unique physical boundaries, a check was performed to see how much the result moved when the cuts were changed.

For the nominal selection:

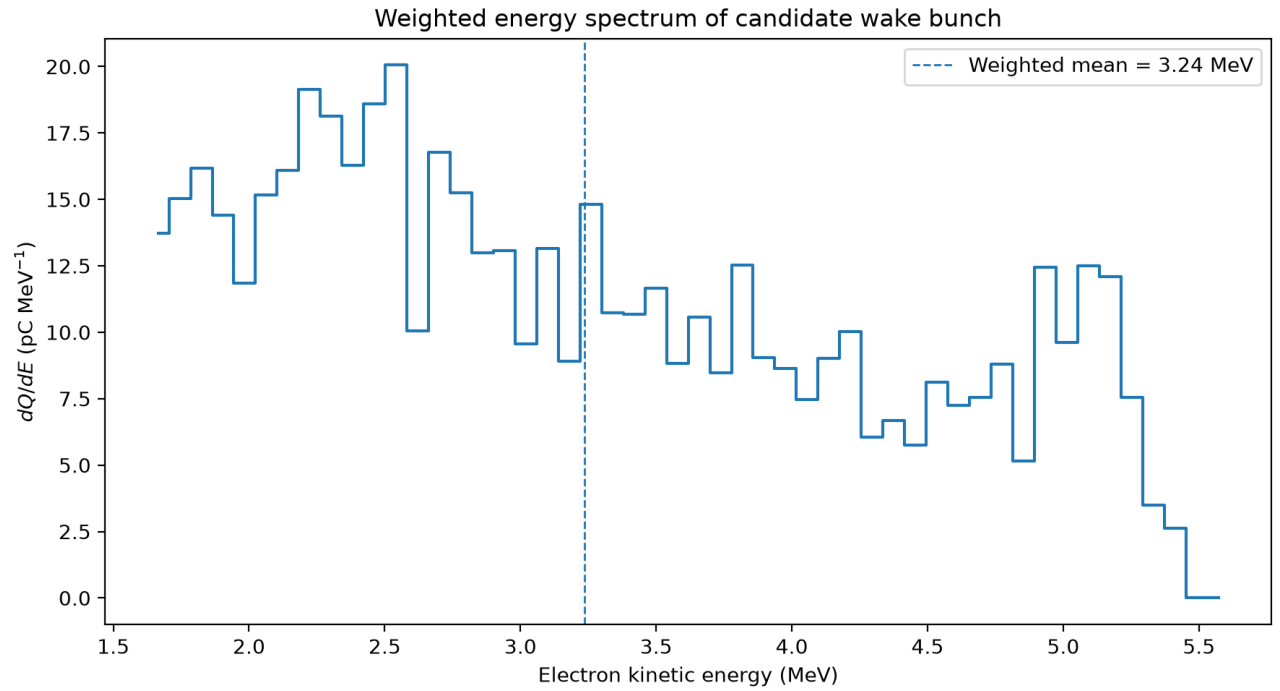
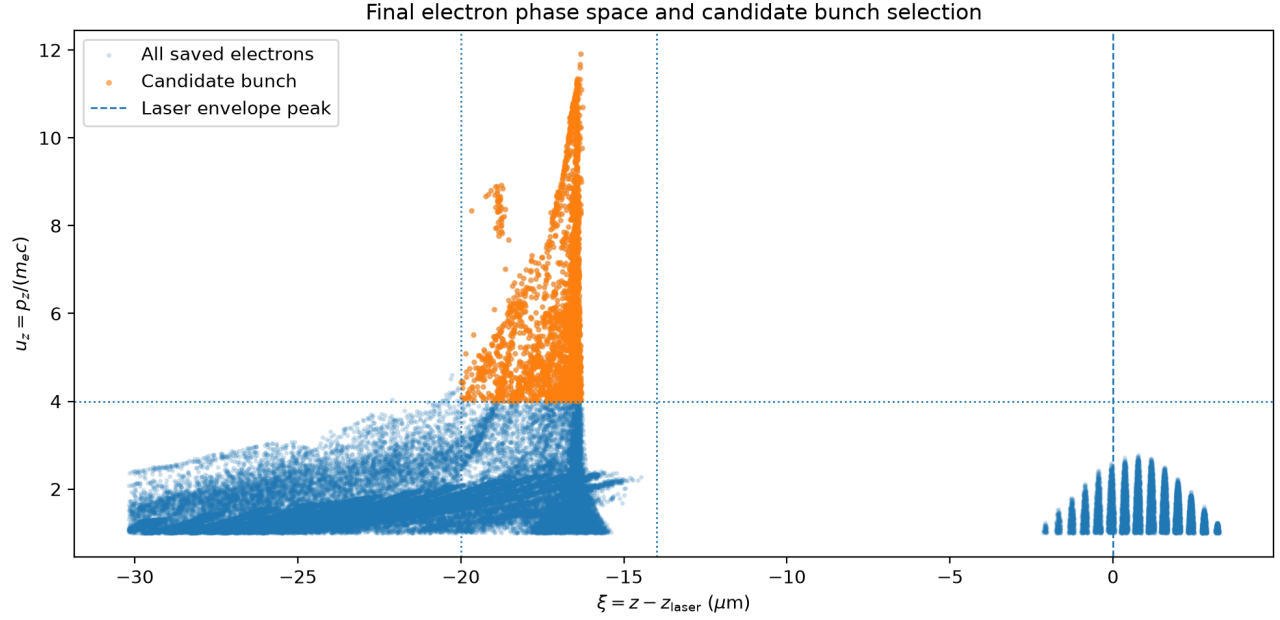
Measurement	Result
Saved macroparticles in full $u_z \geq 1$ diagnostic	50,357
Candidate-bunch macroparticles	2,609
Q_{saved}	346.916 pC
Q_{bunch}	43.254 pC
Fraction of saved charge	12.47%
Weighted mean bunch energy	3.235 MeV
Histogram peak energy	2.542 MeV
Maximum bunch energy	5.611 MeV
Weighted RMS energy spread	1.072 MeV
Relative RMS energy spread	33.14%

The selected population is clearly broad rather than quasi-monoenergetic. The 5.611 MeV value is only the end of the high-energy tail, so I would not describe this as a 5.6 MeV beam. The weighted mean, 3.235 MeV, is a better description of the selected population, and the 33% relative RMS spread shows how broad it still is.

The low-energy edge of the selected spectrum is partly imposed by the $u_z \geq 4$ momentum cut itself. The spectrum should therefore be interpreted together with the phase-space selection rather than as an independent natural low-energy boundary.

Associated files:

- [analysis/analyse_final_bunch.py](#)
- [figures/final_bunch_selection.png](#)
- [figures/final_bunch_energy_spectrum.png](#)



Diagnostic 13.2: candidate-bunch selection sensitivity

The spatial and momentum limits were varied to test whether the candidate-bunch conclusion depended on a narrow choice of selection boundaries.

Spatial-window sensitivity

With $u_z \geq 4$ fixed:

ξ range	Q (pC)	Weighted mean energy (MeV)	Relative RMS spread	Macroparticles
-21 to -14 μm	43.599	3.228	33.18%	2,626
-20 to -14 μm	43.254	3.235	33.14%	2,609
-19 to -14 μm	40.710	3.279	33.24%	2,510
-20 to -13 μm	43.254	3.235	33.14%	2,609
-20 to -15 μm	43.254	3.235	33.14%	2,609

The spatial cut is not especially sensitive. Moving the boundaries by about 1 μm only changes the charge by a few pC, while the mean energy and relative spread hardly move. Shifting the right-hand edge does nothing at all in several cases, which shows that the selected high- u_z branch already ends before that boundary.

Momentum-cut sensitivity

With $-20 \leq \xi \leq -14 \mu\text{m}$ fixed:

Minimum u_z	Q (pC)	Weighted mean energy (MeV)	Relative RMS spread	Macroparticles
3.5	50.433	3.009	37.94%	3,376
4.0	43.254	3.235	33.14%	2,609
4.5	37.208	3.449	28.99%	2,048
5.0	31.316	3.676	25.08%	1,594

The momentum cut matters more, which makes sense because it passes through the lower edge of the branch itself. Increasing the threshold removes more of the lower-energy population, so the selected charge falls while the mean energy rises and the relative spread gets smaller.

Even so, every reasonable cut gives the same general picture: tens of pC, a few MeV and a broad spectrum. A momentum cut-off of $u_z \geq 4$ was kept as the nominal choice because it follows the visible separation in phase space and is close to the 95th percentile of the full saved population. The exact bunch numbers should still be read as selection-dependent.

Associated file:

- [analysis/test_bunch_selection_sensitivity.py](#)

Overall interpretation

The diagnostics now present a clear picture of this first run. At the final snapshot, the laser sits near $z \approx 103 \mu\text{m}$, with the clipped density plot clearly showing a depleted cavity and a dense sheath which closes near $z \approx 86.5 \mu\text{m}$. As this is almost exactly where the strongest negative on-axis E_z appears, these features together provide strong evidence that a wake formed.

The final phase-space plot also puts the high-momentum branch in this trailing region, and separates it from the regular oscillatory structure that moves with the laser. This suggests that the early maximum-energy particles and the final wake-associated branch are not the same population.

One of the main lessons from the baseline is that neither peak E_z nor E_{max} tells the whole story. The wake can strengthen while the maximum saved-particle energy falls, and the full saved spectrum mixes laser-associated and wake-associated electrons. Bunch metrics only become useful after the relevant population has been identified in phase space.

For the nominal late-time bunch selection, the result is approximately

$$Q_{\text{bunch}} \approx 43 \text{ pC},$$

$$\bar{E}_{\text{bunch}} \approx 3.24 \text{ MeV},$$

with a relative RMS energy spread of approximately

$$\frac{\sigma_E}{\bar{E}} \approx 33$$

The exact numbers move with the lower momentum cut, but not enough to change the main conclusion: the late wake-associated population is at the tens-of-pC level, has a characteristic energy of a few MeV and is broad-spectrum rather than cleanly quasi-monoenergetic.

Limitations and next checks

- The initial $16.2 \mu\text{m}$ peak-separation result is not an unambiguous wake-wavelength measurement because the second detected E_z maximum lies within the laser region.
- The exact $127.6n_0$ density spike and -680 GV m^{-1} field minimum are highly localised. Their associated large-scale structures are physically coherent and the E_z spike survives modest spatial averaging, but their exact amplitudes are not yet numerically converged.
- The z - u_z scatter plots are unweighted visualisations. Macroparticle weights are included in charge and energy distributions but not in the displayed scatter-point density.
- The particle diagnostic excludes electrons with $u_z < 1$, so none of the saved-particle plots represent the complete plasma-electron population.
- The candidate bunch is defined using explicit spatial and momentum cuts. The spatial selection is relatively insensitive to small changes, while the inferred charge, mean energy and spread vary more noticeably with the lower momentum threshold.
- The Hilbert-envelope diagnostic measures the position and peak amplitude of the on-axis E_x envelope; it does not by itself measure total laser pulse energy, depletion or spot-size evolution.
- Numerical convergence with respect to longitudinal/radial resolution, macroparticle sampling and retained azimuthal modes has not yet been tested.
- Before parameter optimisation, at least a small numerical-validation comparison should be performed so that optimisation metrics are not dominated by resolution, sampling or mode choices.
- Individual particle IDs were not tracked between saved snapshots, so the phase-space evolution distinguishes changing populations by their structure and position but does not provide individual electron trajectories.